# AI-Assisted E-Learning and Wits EIE Curriculum Recommendation Platform: Project Plan (26P73)

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**Abstract:** This project plan proposes an integrated AI-assisted e-learning and Wits EIE curriculum recommendation prototype. The system will combine a versioned curriculum database, deterministic prerequisite and stream gates, preference extraction, content similarity, hybrid ranking, conversational clarification and feedback reranking. It will support pathway, stream and elective advising alongside a public course library with descriptions, ratings and reviews, learning-journey progress and achievements. Evaluation will use fixed test cases and matched variants to measure eligibility violations, relevance, clarification, source support, refusal behaviour and feedback-loop correctness. Human usability evaluation will follow the approved protocol and University research permission. The expected outcome will be a reproducible, source-grounded prototype and evidence showing whether eligibility gates, personalised ranking, clarification and feedback improve recommendation correctness and usefulness without claiming official registration approval.

**Key words:** academic advising, conversational recommender systems, curriculum database, education technology, recommender systems.

## 1. INTRODUCTION

Engineering students must make pathway, stream and elective choices while interpreting prerequisites, curriculum rules and module descriptions. These choices affect later module availability, but the supporting information is distributed across rules documents, syllabuses, course briefs and school guidance.

The engineering problem is therefore not simply information retrieval. Useful advice must be relevant to a student's interests, consistent with curriculum constraints and auditable against source evidence. A pure retrieval-augmented-generation system is unsuitable as the core decision engine because prerequisite and stream rules must be enforced as hard constraints, not inferred from retrieved text.

### *1.1 Literature Review*

Recommender systems reduce information overload by ranking items likely to be useful to a user [1, 2]. Collaborative filtering and matrix factorisation are effective when large interaction histories are available [3], but are not a good primary fit here: no large ethics-cleared dataset of Wits choices and ratings is available, private academic records should not be used, and an invalid prerequisite recommendation is more serious than an irrelevant product recommendation. This limitation concerns collaborative filtering from Wits interaction histories; it does not exclude transparent use of public aggregate ratings, public reviews or session-only feedback as secondary course-ranking signals.

Knowledge-based and constraint-based recommenders are better suited because they use explicit item knowledge, user preferences and domain constraints [4, 5]. Course recommendation under prerequisite and requirement constraints has also been studied directly, showing that constraints need dedicated handling rather than ordinary ranking alone [6]. Hybrid systems remain useful after hard rules have been applied because they combine rule, content and preference evidence [7, 8].

Educational recommenders add learner context, curriculum structure and pedagogical goals [9, 10]. A recent review confirms higher-education uses such as course selection and e-learning support, but a local curriculum-rule-grounded advisor remains a narrower engineering problem [11]. Conversational recommender systems extend one-shot lists by asking clarifying questions, carrying dialogue state and allowing users to refine preferences [12]. Explanation work supports showing why an item was recommended and which evidence was used [13].

Retrieval-augmented generation highlights provenance for knowledge-intensive responses [14]. Here, source grounding means traceable curriculum evidence, not a RAG architecture: recommendations will remain knowledge- and constraint-based so prerequisite and stream rules stay as hard gates. Recommender evaluation must also consider coverage, robustness, constraint satisfaction and user-facing usefulness [15]; controlled offline comparison is established practice for recommender variants [16].

### *1.2 Research Gap and Investigation Question*

The gap addressed is the combination of four constraints: prerequisite and stream rules must be hard gates, curriculum claims must trace to local Wits sources, preference elicitation must work without private histories, and conversational clarification must avoid unsupported official-advice claims. The investigation will isolate these design elements with matched

variants rather than assume that a final hybrid design is superior.

This project will implement the assigned AI-assisted e-learning recommendation platform [17] as an integrated course platform and Wits EIE pathway/module advisor governed by the 2026 curriculum rules [18]. The central investigation question is:

*Do hard eligibility gates, personalised ranking and conversational clarification improve the correctness and usefulness of Wits EIE curriculum recommendations compared with matched simpler strategies?*

The prototype will be experimental rather than an official registration tool. It will use public or project-supplied information and synthetic/session-only profiles, not private records, marks, financial information or persistent participant chat logs.

## 2. AIM AND OBJECTIVES

The aim is to design, implement and evaluate a reproducible e-learning and curriculum recommendation prototype. The sub-questions test whether hard gates eliminate prerequisite and stream violations, whether personalised ranking improves relevance without reintroducing violations, whether clarification improves ambiguous-prompt answers, and whether session feedback improves later course ranking without changing curriculum rules.

The objectives are to:

1. build versioned curriculum and course databases with source trails;
2. use synthetic/session-only profiles containing goals, interests and declared learning-format preferences;
3. provide course descriptions, ratings, reviews, journey progress and achievements;
4. apply prerequisite, stream, exclusion and source-verification gates;
5. implement preference extraction, content matching and personalised ranking of eligible options;
6. support conversational clarification, explanations, source trails and out-of-scope refusal;
7. use ratings, critiques and progress feedback to rerank only eligible options;
8. compare four matched variants under identical data, profiles and prompts;
9. validate databases, recommendations, dialogue and feedback using automated and statistical tests; and
10. conduct approved usability evaluation with up to 30 adult Wits students.

## 3. PROPOSED SYSTEM DESIGN

The prototype will use a local deterministic architecture rather than a live large-language-model service. This supports reproducibility, simplifies testing and reduces the risk of sending participant chat content externally. A Streamlit interface with local CSV/JSON files and a SQLite database is sufficient for a school-scale prototype, where traceability and repeatable validation are more important than production deployment.

### *3.1 System Architecture and Data Model*

Table 1 : Planned system architecture.

| Layer | Responsibility |
|---|---|
| Source | Official/public or supplied rules, syllabuses, colour-book guidance and module information. |
| Data | Structured modules, rules, courses, profiles, ratings, reviews, progress, achievements and source trails. |
| Advisor | Intent detection, state tracking, hard gates, hybrid scoring, clarification and grounded explanations. |
| Interface | Pathway/stream guidance, module/elective recommendations, course library, ratings/reviews, journey, progress, achievements, feedback, chat and evidence views. |
| Validation | Database integrity, prerequisite consistency, chat regression, variant comparison and app smoke tests. |

The database will cover the relevant 2026 Wits EIE pathways, streams, modules and elective rules, and at least 20 verified public courses spanning programming/data/AI, electronics/control, communications and career skills, beginner/intermediate levels and at least three providers. Every material curriculum rule will have a source reference.

Table 2 : Indicative database schema.

| Record | Main fields |
|---|---|
| Module | Code, title, stream/group, prerequisites, co-requisites, exclusions, outcomes, topic/interest/career tags, workload, format and source ID. |
| Course | ID, title, provider, category/topic/interest/career tags, outcomes, workload, format and rating/review statistics. |
| Rule | Item, rule type, related item, applicability/status and source ID. |
| Source | Document, year, section, page and access date. |
| Profile | Stage/stream, completed/current/planned modules, interests, career preference, workload and learning-format preference. |
| Session | Current intent, prior choices, clarification state and eligible set. |
| Alias | Alias text, canonical tag and tag type. |
| Feedback | Item ID, rating, critique signal, category tags and session timestamp. |

The profile will contain declared interests, completed/current/planned modules, career goals, workload and learning-format preference, not psychological, demographic or private records. Given the weak evidence for matching instruction to fixed learning styles [19], format will be a disclosed low-weight preference only. Source authority will be: current Faculty rules, official School guidance, official module outlines, then verified public sources. Conflicts will be flagged for manual verification. Interest, career and alias tags will use a versioned controlled vocabulary. Both students will tag and source items independently, reconcile disagreements, record the rationale and freeze mappings before variant outputs are inspected.

### 3.2 Eligibility and Recommendation Logic

A prompt will be normalised, mapped to an advising stage and matched to preference areas such as programming, control, telecommunications, power, data systems or workload. Ineligible options will be removed before ranking. For example, ELEN4001A will be excluded for an IE profile because the 2026 rules list it only in the EE elective pool; for an EE profile it remains ineligible unless ELEN3000A and MATH3025A are satisfied.

Prerequisite, co-requisite, stream, exclusion and source-verification gates will be applied before ranking; an ineligible or materially unsupported option will not be recommended regardless of score. Eligible option $i$ will initially be ranked by

$$S_i = 0.33I_i + 0.28C_i + 0.22T_i + 0.17W_i, \qquad (1)$$

where $I_i$, $C_i$, $T_i$ and $W_i$ are normalised $[0, 1]$ interest, career, TF–IDF content-similarity and workload/format matches. The heuristic weights will be frozen before outputs are inspected and varied by $\pm 20\%$, one at a time, to test ranking sensitivity. Each run will multiply one nominal weight by 0.8 or 1.2, renormalise all four by their new sum and hold data, prompts, gates, seed and other parameters fixed. Evidence completeness will be an independent integrity gate/metric, not a personal-suitability signal. Appendix E specifies component calculations, preprocessing and missing-value handling.

### 3.3 Conversational, Journey and Feedback Functions

The advisor will ask one focused question when a required field is missing, a request is ambiguous/conflicting, the top-score margin is below 0.05, or no preference match exceeds 0.40; unrelated prompts will be redirected. Each recommendation will include an explanation, disclaimer, source trail and any uncertainty warning. Course progress and completed activities will update session achievements. Feedback will rerank only eligible options using

$$S'_i = (1 - \beta)S_i + \beta F_i, \qquad \beta = 0.10, \qquad (2)$$

where $F_i$ is normalised session feedback. Without feedback, $S'_i = S_i$; feedback will not modify curriculum rules or either source database.

### 3.4 Experimental Variants

The matched variants will be: (1) eligibility gates followed by a fixed non-personalised order; (2) ungated personalised ranking; (3) the identical personalised score with gates; and (4) the identical gated score with clarification. Thus, ungated–gated isolates eligibility, eligibility-only– gated-personalised isolates ranking, and gated-personalised–clarification isolates dialogue. Data, profiles, prompts, weights and scoring rubrics will be identical wherever applicable. Appendix E also fixes the gate order, clarification policy, minimum interface and acceptance cases so implementation does not require undocumented choices.

## 4. IMPLEMENTATION PLAN

Implementation will proceed from data integrity to recommendation behaviour and user-facing evaluation. The curriculum advisor, course platform, journey and feedback loop are core scope; visual refinements will follow functional validation. Table 3 provides the work breakdown and schedule by linking each implementation phase to its evidence output, as required by the project CBO [20].

Table 3 : Work breakdown, eight-week schedule and outputs.

| Period | Main work | Output |
|---|---|---|
| Week 1 | Finalise scope, data boundaries, permissions and schema. | Requirements and risks. |
| Week 2 | Build curriculum/course data, sources, ratings and rules. | Draft databases. |
| Week 3 | Audit records, rule consistency and ambiguous sources. | Validated data. |
| Week 4 | Implement gates, ranking, clarification, feedback and variants. | Tested engine. |
| Week 5 | Integrate pathway, course library, journey, achievements, chat and evidence views. | Working prototype. |
| Week 6 | Run automated tests; freeze reference cases and prepare approved survey distribution. | Validation results. |
| Week 7 | Compare variants and analyse permitted usability, sources, failures and uncertainty. | Evaluation evidence. |
| Week 8 | Complete poster, two individual reports, presentation and evidence pack. | Final artefacts. |

Resources are ordinary laptop development tools, Python, Streamlit, local CSV/JSON/SQLite artefacts, LaTeX, version control and official/public curriculum documents. No laboratory equipment, specialised hardware, paid cloud service or additional project budget is required. Gift Vundla will lead advisor integration, validation and release packaging; Vukosi Khoza will lead curriculum/course data, intent/clarification and the Streamlit interface. Both will jointly manage the project, freeze reference cases, score blinded outputs, analyse results and prepare the final deliverables, as detailed in Appendix D.

## 5. EVALUATION AND ETHICS

The evaluation will test whether the prototype is correct, useful and defensible.

### 5.1 Offline Validation

Database checks will test module-code uniqueness, prerequisite consistency, stream gates, source trails, descriptions and references. Recommendation tests will verify hard constraints and detect invented module codes or prerequisites. Course-platform tests will verify course-record integrity, rating/review link-

age, journey and achievement state, feedback-based reranking and that session feedback does not change either database. Thirty to fifty scripted profiles will generate at least 150 fixed cases across stage, eligibility, vague/conflicting interests, prerequisites, comparisons, typos, topic changes and unrelated prompts. Expected answers and the rubric will be frozen before outputs are inspected. A 10–15 case anchor subset will be independently checked against official sources and, where permitted, reviewed by an academic advisor.

### *5.2 Comparative Evaluation*

The four matched variants in Section 3.4 will use identical data and test cases. The ungated variant is expected to produce eligibility violations, while gated variants should produce zero. Personalisation is expected to improve relevance over fixed ordering, and clarification should improve ambiguous-prompt correctness at the cost of longer interaction.

Outputs will be randomised and variant labels hidden during scoring. Both students will score independently. Linearly weighted Cohen's kappa will summarise ordinal ratings, while percentage exact agreement will be reported for binary decisions. Each paired comparison will use the profile-level mean metric difference and a two-sided percentile 95% interval from 10,000 paired cluster-bootstrap resamples of profiles with replacement, seed 26073, retaining all prompts per sampled profile; intervals containing zero will be inconclusive. Ranking weights and clarification thresholds will also be varied by ±20%, one at a time.

Table 4 : Evaluation plan.

| Question | Metric and evidence |
|---|---|
| Database | Integrity, source-trail and prerequisite audits. |
| Eligibility | Unknown-code, prerequisite and stream-violation rates; targets 0% for gated outputs. |
| Ranking | Precision@3 target $\geq 0.80$; mean relevance target $\geq 4/5$. |
| Clarification | Appropriateness and post-clarification correctness target $\geq 90\%$. |
| Grounding | Source support for material curriculum claims; target 100%. |
| Boundary | Out-of-scope refusal/redirection target $\geq 90\%$. |
| Course platform | Course/review integrity, journey state and feedback reranking. |
| Usability | Anonymous clarity, usefulness, trust and explanation ratings. |

Relevance will be scored 1–5: 1 unsuitable/constraint-mismatched; 2 weak fit with a major preference mismatch; 3 partly useful with mixed fit; 4 useful with only a minor mismatch; and 5 a strong fit to all stated constraints. Tests and targets will not change after outputs are known. Precision@3 will be calculated only with at least three eligible candidates. Otherwise, top-1 is correct when the first item belongs to the frozen acceptable set; exact-set correctness requires equality of predicted and frozen sets and ignores order unless ranking is the tested outcome.

Clarification-trigger appropriateness will use all clarification test cases; post-clarification correctness will use only cases receiving a valid follow-up answer. Abandoned/invalid follow-ups will be reported separately. Student raters will know the hypotheses despite hidden labels; a permitted reviewer will inspect a blinded subset and residual investigator bias will be reported. Usability results will be exploratory, summarised by $n$, distributions, median and interquartile range, without population-level inference.

### *5.3 Ethics*

The protocol is approved unconditionally as minimal risk under ELEN-UG-26E02. University research permission was granted on 15 July 2026 and expires on 15 December 2026. Up to 30 adult Wits students may participate voluntarily and anonymously, may withdraw at any time, and will receive the approved information and consent material. No names, student numbers, marks, health, financial or demographic data, or persistent chat logs will be collected; results will be aggregate. Before recruitment, the team will provide the prototype/survey link, invitation email, target faculty and study years. If specific classes, courses or cohorts are approached during teaching, the relevant Head of School or lecturer permission will also be obtained.

## 6. MILESTONES, RISKS AND DELIVERABLES

The core deliverables are validated curriculum and course databases, eligibility and personalised-ranking engines, conversational advisor, course library, journey/progress/achievement tracking, ratings and feedback, four matched variants, automated tests, offline and permitted usability results, two individual reports, poster, presentation and reproducibility/evidence pack.

The main risks are ethics-scope drift, unsupported curriculum claims, source errors, delayed survey distribution or cohort permission, biased reference cases, inconclusive variants, scope growth, late deliverables, unequal work and data-security weaknesses. Controls include source-level verification, frozen cases, fixed tests, confidence intervals, staged integration, weekly records, backups and early internal deadlines. Restricted sources will not be redistributed; participant chat logs and sensitive records will not be stored. OHS risk is low because the work is software-only and desk/online based.

## 7. CONCLUSION

The planned investigation will produce an integrated e-learning and Wits EIE curriculum-recommendation prototype with eligibility-safe, source-grounded and personalised recommendations, conversational clarification, course-library, progress, achievement and feedback functions. Its evidence will test which com-

ponents improve correctness and usefulness against matched baselines, without claiming official registration authority.

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# APPENDICES

## A PROJECT WORK BREAKDOWN STRUCTURE

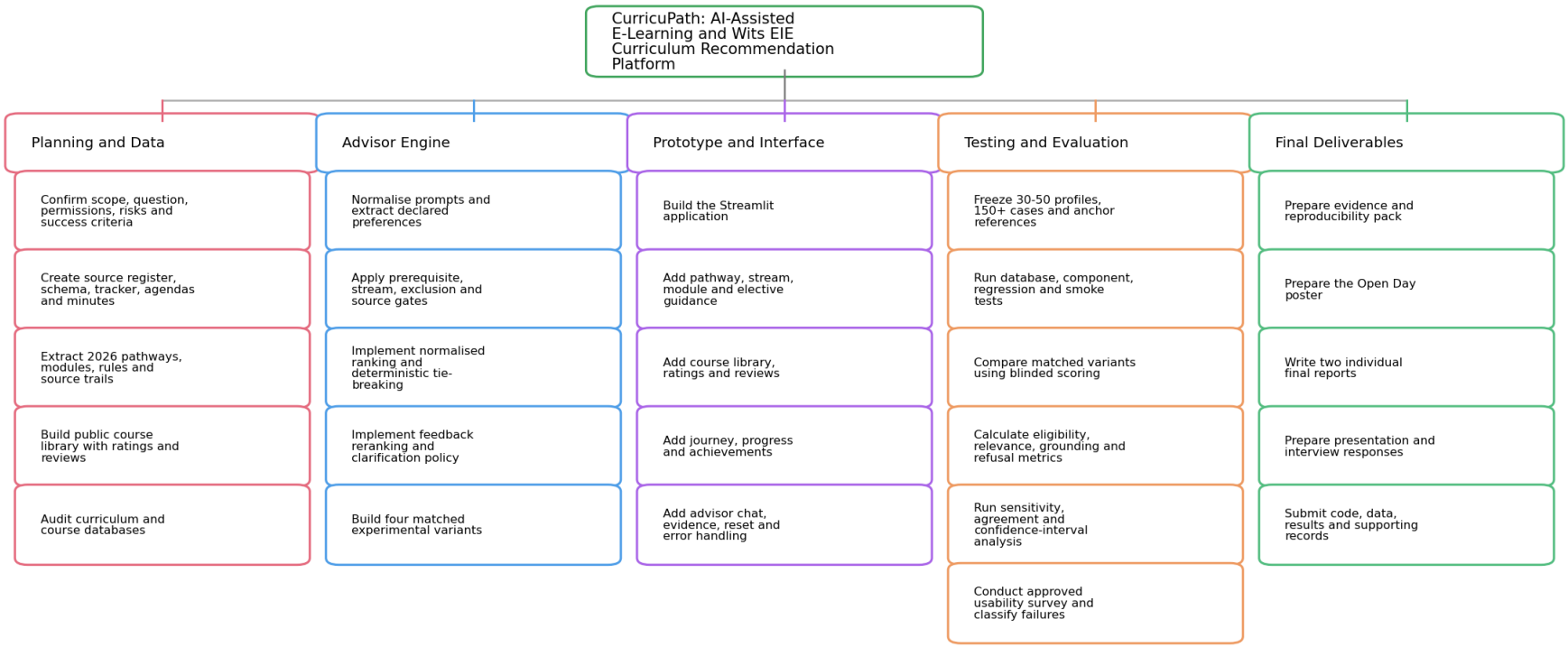



Figure A.1 : Project work breakdown structure. "CurricuPath" is the working label for the integrated e-learning and Wits EIE curriculum recommendation platform.

## B PROJECT GANTT CHART

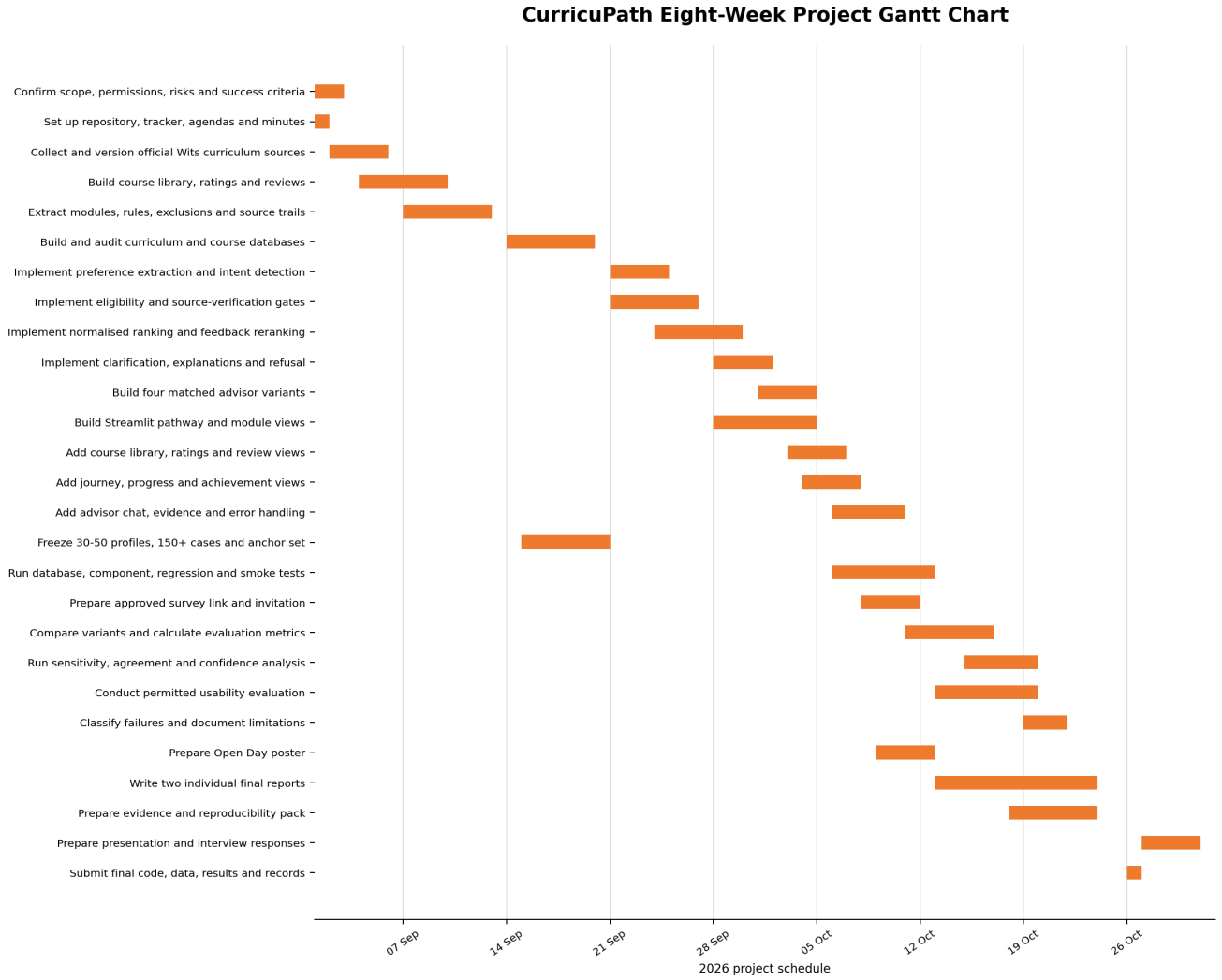



Figure B.1 : Eight-week project Gantt chart covering the integrated course platform, curriculum advisor, evaluation and individual reports.

## C RISK REGISTER

The register will be reviewed during weekly project meetings.

Table C.1 : Project risk register.

| ID | Risk | Likelihood | Impact | Level | Mitigation |
|---|---|---|---|---|---|
| R1 | Incorrect or outdated curriculum data | Medium | High | High | Use current official sources, record versions and audit all prerequisite and stream rules. |
| R2 | Scope exceeds the eight-week period | Medium | High | High | Prioritise databases, gates, four variants, core interface and evaluation. |
| R3 | An ineligible option is recommended | Low | High | Medium | Apply deterministic gates before ranking and require zero gated violations. |
| R4 | Implementation or integration delay | Medium | Medium | Medium | Develop incrementally, test early and retain a working minimum prototype. |
| R5 | Reference cases are biased or incorrect | Medium | Medium | Medium | Freeze source-grounded expected sets before inspecting variant outputs. |
| R6 | Independent review is unavailable | Medium | Medium | Medium | Seek permitted review early; retain source checks and report the limitation. |
| R7 | Survey distribution is delayed | Medium | Medium | Medium | Prepare the approved link and invitation early; preserve the offline investigation. |
| R8 | Additional cohort permission is required | Low | Medium | Low | Obtain Head of School or lecturer permission before targeting a class or year group. |
| R9 | Variant differences are inconclusive | Medium | Medium | Medium | Use fixed cases and confidence intervals; do not overstate small effects. |
| R10 | Curriculum sources conflict | Medium | High | High | Apply the authority hierarchy and manually verify unresolved interpretations. |
| R11 | Code, data or results are lost | Low | High | Medium | Use Git, shared backups, regular commits and archived releases. |
| R12 | Sensitive information is collected | Low | High | Medium | Exclude marks, identifiers, academic records, demographics and persistent chat logs. |
| R13 | IP or licence infringement occurs | Low | Medium | Low | Cite sources and do not redistribute restricted documents. |
| R14 | Poster or reports are late | Medium | High | High | Use internal deadlines before the official submission dates. |
| R15 | Group contribution becomes unequal | Low | Medium | Low | Assign ownership, cross-review work and record contributions in notebooks and minutes. |
| R16 | Workstation-related OHS incident occurs | Low | Low | Low | Follow normal ergonomic and electrical-workstation precautions. |

## D STUDENT WORK SPLIT

Table D.1 : Planned student work split.

| Work package | Primary lead | Main responsibilities | Joint responsibilities |
|---|---|---|---|
| Planning and management | Both | Scope, schedule, meetings, risks and records. | Review progress and approve changes. |
| Curriculum data and sources | Vukosi Khoza | Extract modules and rules, structure data, record source trails and ambiguities. | Audit records and verify interpretations. |
| Advisor engine and integration | Gift Vundla | Integrate gates, ranking, variants, session logic and reproducible configuration. | Review design and interfaces. |
| Intent and clarification | Vukosi Khoza | Implement normalisation, intent mapping, preference extraction and follow-up policy. | Review prompts and failure handling. |
| Prototype and course platform | Vukosi Khoza | Develop Streamlit views, course library, ratings, journey, achievements, feedback, chat and evidence interfaces. | Test usability and end-to-end behaviour. |
| Validation and release | Gift Vundla | Implement database, variant, feedback-loop, regression and smoke tests; package releases. | Review coverage and reproduce final runs. |
| Evaluation | Both | Freeze expected sets, score blinded outputs, analyse variants and classify failures. | Independent scoring and agreement checks. |
| Documentation | Both | Prepare poster, two individual reports, presentation, evidence pack and viva material. | Cross-review technical accuracy. |

## E IMPLEMENTATION RULES AND ACCEPTANCE CRITERIA

These deterministic rules are part of the hand-off specification. They will be frozen before formal testing; changes will be versioned and justified.

Table E.1 : Eligibility and source-gate order.

| Order | Gate | Required action |
|---|---|---|
| 1 | Source verification | Require a current authoritative source for material rules; flag disputed or incomplete records for manual verification and do not recommend them. |
| 2 | Programme/stream | Retain only pathways/modules permitted for the selected programme and stream. |
| 3 | Prerequisite | Require every recorded prerequisite in the completed-module set; unknown completion status triggers clarification. |
| 4 | Co-requisite | Require each recorded co-requisite in the completed or current/planned set, as permitted by the source rule; unknown current selection triggers clarification. |
| 5 | Exclusion | Remove prohibited combinations before calculating suitability. |
| 6 | Ranking | Score and order only the candidates that passed Gates 1–5. |

Table E.2 : Recommendation-component calculations.

| Term | Range | Fixed implementation rule |
|---|---|---|
| $I_i$ | $[0, 1]$ | Size of the set intersection between supplied user-interest tags and documented item-interest tags, divided by the number of unique supplied tags. Duplicate tags are removed. |
| $C_i$ | $[0, 1]$ | Size of the set intersection between supplied user-career tags and curated item-career tags, divided by the number of unique supplied tags. Duplicate tags are removed. |
| $T_i$ | $[0, 1]$ | Cosine similarity between TF–IDF vectors for combined profile/query text and item title, description, outcomes and topic tags. Text is lower-cased; punctuation and English stop words are removed; unigram/bigram features are used. One vectoriser is fitted once on the frozen module/course corpus; all prompts and variants use that vocabulary without refitting. |
| $W_i$ | $[0, 1]$ | Arithmetic mean of available workload and format sub-scores. Workload uses light/moderate/heavy (same/adjacent/opposite = 1/0.5/0). Format uses present/unspecified/absent = 1/0.5/0 only when the user states a preference; unspecified means verified item metadata is missing. If exactly one sub-score is available, use it. |
| Missing personalisation | – | After one clarification attempt, omit $W_i$ only if both sub-scores are unavailable; omit any other unresolved personalisation component and proportionally renormalise the remaining weights. Missing user format preference follows this clarification/omission rule. |
| Order | – | Sort by descending score; resolve exact ties by ascending module/course code, or alphabetically by title when no code exists. |

Table E.3 : Course, feedback and journey rules.

| Function | Fixed implementation rule |
|---|---|
| Feedback $F_i$ | Convert a 1–5 rating to $(r-1)/4$; map helpful/neutral/unhelpful critique signals to 1/0.5/0. Average available direct-item signals; otherwise average signals from matching course-category tags. Direct item feedback takes precedence. |
| Behaviour | Only explicit session actions affect $F_i$: save, helpful or confirmed completion maps to 1; an explicit neutral response to 0.5; and skip or unhelpful to 0. No action creates no signal. Direct ratings take precedence. Passive clickstreams and persistent event histories are prohibited. |
| Reranking | Apply $S_i' = (1 - 0.10)S_i + 0.10F_i$ only to eligible candidates. With no feedback, retain $S_i$. Never change source rules or stored course/curriculum records. |
| Ratings/reviews | A public record is source-verified only when provider/platform URL, access date, scale, mean $\mu_p$ and count $n_p$ are stored. For compatible 1–5 ratings display $(n_p\mu_p + \sum r_s)/(n_p + n_s)$, total count and separate public/session counts. Exclude duplicates and incompatible scales; one session rating per item may be replaced but is not persisted. |
| Journey/achievement | Viewing content does not change progress. Progress changes only after an explicit completion action; award an achievement only when its predefined course activities or advising milestone are complete. |

*Clarification and Interaction Rules*

The policy will resolve eligibility-critical fields first (stage, stream, prerequisite completion and current/planned co-requisites), then select the unresolved field expected to distinguish the leading eligible candidates most. One template-based question will be asked per request. Its answer will be normalised with the alias tables, written to session state, and gates and ranking will be rerun. If an eligibility-critical field remains unresolved, the advisor will make no recommendation and return an insufficient-information message. If only personalisation remains unresolved, it may return verified eligible options in fixed non-personalised order with an uncertainty warning.

Table E.4 : Clarification decision table.

| Condition | System action | Example question/response |
|---|---|---|
| Stage/programme missing | Ask for year and programme/stream. | "Are you choosing a first-year pathway, EE/IE stream or fourth-year electives?" |
| Relevant completion unknown | Ask only about candidate prerequisites. | "Have you completed ELEN3000A and MATH3025A?" |
| Current co-requisite unknown | Ask whether the source-listed co-requisite is completed or in the current/planned selection. | "Have you completed, or do you plan to take, the required co-requisite concurrently?" |
| Interest/career missing | Ask for one preferred area or career direction. | "Which matters most: control, communications, power, programming or data?" |
| Workload missing and discriminating | Ask for light, moderate or heavy preference. | "Which workload level do you prefer?" |
| Top-two margin $< 0.05$ | Ask about the feature that most separates the two. | "Would you prefer more programming or communications theory?" |
| Best preference match $< 0.40$ | Request one additional subject or skill. | "Name one skill the recommendation should emphasise." |
| Conflicting answers | Ask which stated preference has priority. | "Which matters more: a lighter workload or intensive practical work?" |
| Eligibility still unresolved | Do not recommend an item; state that eligibility cannot be established. | "I need the missing programme or module-status information before recommending." |
| Personalisation still unresolved | Return verified eligible options in fixed order with a warning. | "These are eligible options, not a confident personal ranking." |
| Unrelated request | Redirect to supported project scope. | "I can help with Wits EIE pathways, modules and related learning courses." |

Table E.5 : Minimum interface and acceptance cases.

| Area | Minimum implementation | Acceptance evidence |
|---|---|---|
| Interface | Profile setup; curriculum advisor; course library/details/reviews; journey; achievements; chatbot; evidence/source view. | Every view loads, preserves only session state, exposes reset/error handling and contains no sensitive-field input. |
| Eligibility | IE profile requests ELEN4001A; EE profile lacks MATH3025A. | ELEN4001A is excluded for IE and for the incomplete EE profile, with the applicable reason/source. |
| Co-requisite | A source permits concurrent enrolment and the required co-requisite is planned but not completed. | Retain the candidate; clarify or exclude it if neither completion nor concurrent selection can be established. |
| Clarification | Prompt "I like ML" has no advising stage. | Ask one stage question before ranking; after a valid answer, update state and rerun gates/ranking. |
| Feedback | User marks a recommended public course helpful. | Only eligible course order may change; curriculum/course database hashes remain unchanged. |
| Journey | User opens a course, then explicitly marks all defined activities complete. | Opening causes no progress; completion updates progress and awards only the defined achievement. |
| Boundary | User asks an unrelated question. | Redirect without inventing academic advice and preserve prior advising context. |
| Grounding | A candidate has no verified material-rule source. | Do not recommend it; flag the record for manual verification. |
| Experiment | Run all four variants on the frozen cases. | Data, prompts, scores and weights match; only the named gate/ranking/clarification component changes. |